

JAVA WEB

Unit 4: Java Server Page



Introduction

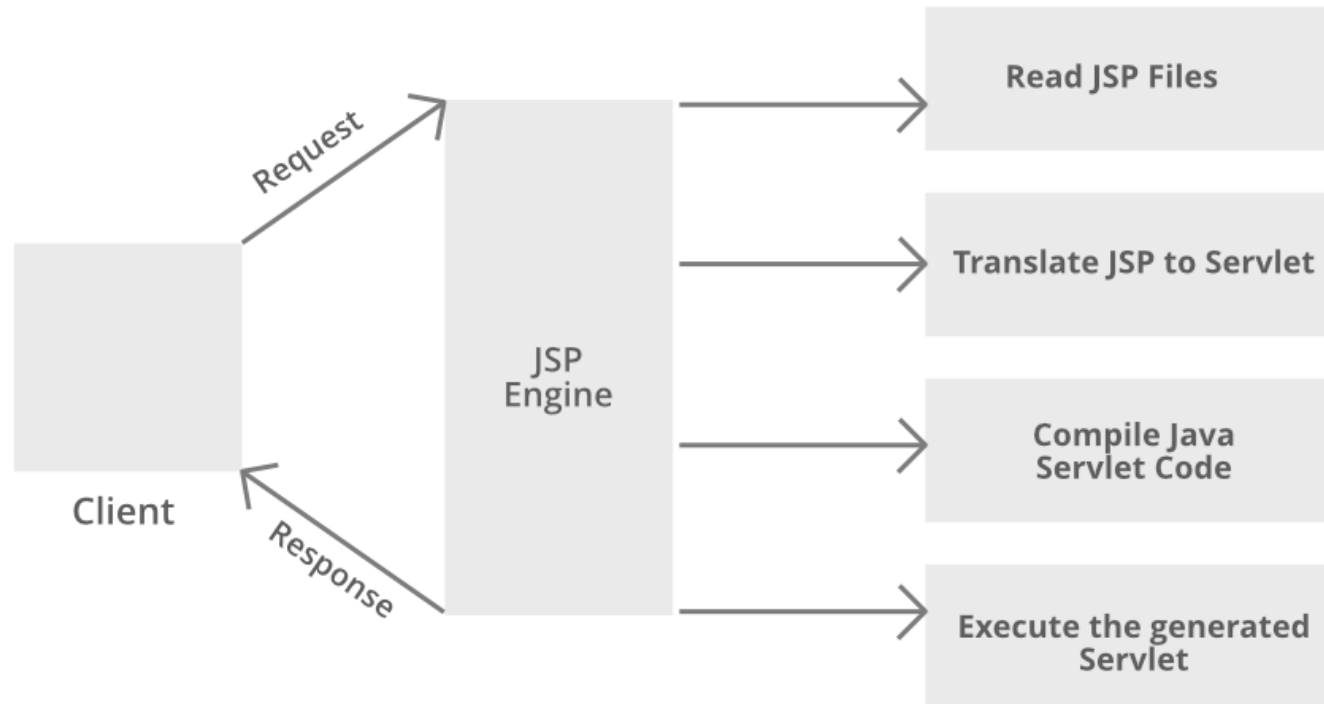
1. **Giới thiệu**
2. **Các phần tử JSP:** biểu thức, đoạn code, khai báo.
1. **Các đối tượng ngầm định:** out, request, response, config, application, session, pageContext, page, exception.
3. **Các thẻ chỉ dẫn:** page directive, include directive, **taglib directive**
4. **Các thẻ hành động:** jsp:forward ; jsp:include;
5. **Java Bean:** jsp:useBean, jsp:setProperty; jsp:getProperty
6. **Xây dựng ứng dụng theo MVC**
7. **Thao tác với CSDL**
8. **Xây dựng ứng dụng E-Commerce minh họa.**

Giới thiệu

JSP is a solution:

- ✓ JSP là một ngôn ngữ script chạy trên **server** được phát triển bởi Sun Microsystems.
- ✓ JSP cho phép **nhúng** mã Java với HTML.

Luồng xử lý:



So sánh Servlet và JSP

JSP

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0 Transitional//EN">
<HTML>
<HEAD>
<TITLE>Reading Three Request Parameters</TITLE>
<LINK REL=STYLESHEET
      HREF="JSP-Styles.css"
      TYPE="text/css">
</HEAD>
<BODY>
<H1>Reading Three Request Parameters</H1>
<UL>
  <LI><B>param1</B>: <%= request.getParameter("param1") %>
  <LI><B>param2</B>: <%= request.getParameter("param2") %>
  <LI><B>param3</B>: <%= request.getParameter("param3") %>
</UL>
</BODY></HTML>
```

Servlet

```
public void doGet(HttpServletRequest request,
                  HttpServletResponse response)
    throws ServletException, IOException {
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();
    String title = "Reading Three Request Parameters";
    String docType =
        "<!DOCTYPE HTML PUBLIC \",-//W3C//DTD HTML 4.0 \" +
        \"Transitional//EN\">\n";
    out.println(docType +
        "<HTML>\n" +
        "<HEAD><TITLE>" + title + "</TITLE></HEAD>\n" +
        "<BODY BGCOLOR=\"#FDF5E6\">\n" +
        "<H1 ALIGN=\"CENTER\">" + title + "</H1>\n" +
        "<UL>\n" +
        "  <LI><B>param1</B>: "
        + request.getParameter("param1") + "\n" +
        "  <LI><B>param2</B>: "
        + request.getParameter("param2") + "\n" +
        "  <LI><B>param3</B>: "
        + request.getParameter("param3") + "\n" +
        "</UL>\n" +
        "</BODY></HTML>");
}
```

Biểu thức

Ví dụ:

```
<html>
<body>
<%= "welcome to jsp" %>
</body>
</html>
```

File: index.jsp

```
<html>
<body>
<form action="welcome.jsp">
<input type="text" name="uname"><br/>
<input type="submit" value="go">
</form>
</body>
</html>
```

File: welcome.jsp

```
<html>
<body>
<%= "Welcome " + request.getParameter("uname") %>
</body>
</html>
```

Đoạn code

<% Java Code %>

JSP/Servlet Correspondence

Listing 11.6 Sample JSP Expression/Scriptlet

```
<H2>foo</H2>  
<%= bar() %>  
<% baz(); %>
```

Listing 11.7 Representative Resulting Servlet Code: Expression/Scriptlet

```
public void _jspService(HttpServletRequest request,  
                        HttpServletResponse response)  
    throws ServletException, IOException {  
    response.setContentType("text/html");  
    HttpSession session = request.getSession();  
    JspWriter out = response.getWriter();  
    out.println("<H2>foo</H2>");  
    out.println(bar());  
    baz();  
    ...  
}
```

Khai báo

<%! **Field** or Method Definition %>

index.jsp

```
<html>
<body>
<%! int data=50; %>
<%= "Value of the variable is:"+data %>
</body>
</html>
```

index.jsp

```
<html>
<body>
<%!
int cube(int n){
return n*n*n*;
}
%>
<%= "Cube of 3 is:"+cube(3) %>
</body>
</html>
```

Các đối tượng ngầm định

```
<html>
<body>
  <form action="welcome.jsp">
    <input type="text" name="username">
    <input type="submit" value="Submit"><br>
  </form>
</body>
</html>
```

//welcome.jsp

```
<html> |
  <body>
    <%
      String name = request.getParameter("username");

      // out.print("welcome " + name);

      response.sendRedirect("http://www.google.com/search?q="+name);
    %>
  </body>
</html>
```


Các đối tượng ngầm định

application Lưu trữ thông tin về ứng dụng.

session Lưu trữ thông tin về phiên làm việc

page lưu thông tin về trang hiện thời, page là tương đương với đối tượng **this**.

pageContext

pageContext là một đối tượng ngầm định, có thể thiết lập phạm vi: **PAGE_SCOPE**, **REQUEST_SCOPE**, **SESSION_SCOPE** và **APPLICATION_SCOPE**. Ví dụ:

```
<html>
<body>
<form action="pageContextDemo.jsp">
<input type="text" name="uname">
<input type="submit" value="go"><br/>
</form>
</body>
</html>
```

//pageContextDemo.jsp

```
<html>
```

```
<body>
```

```
<%
```

```
String name=request.getParameter("uname");
```

```
pageContext.setAttribute("user",name,PageContext.SESSION_SCOPE);
```

```
String nameStr=(String)pageContext.getAttribute("user",PageContext.SESSION_SCOPE);
```

```
out.print("Hello "+nameStr);
```

```
pageContext.removeAttribute("user", PageContext.SESSION_SCOPE);
```

```
String nameStr2=(String)pageContext.getAttribute("user",PageContext.SESSION_SCOPE);
```

```
out.print("Hello "+nameStr2);
```

```
%>
```

```
</body>
```

```
</html>
```

Thẻ chỉ dẫn `<%@page ...>`

Cú pháp: `<%@ page attribute="value" %>`

Attributes of JSP page directive:

- ✓ **contentType:** Định nghĩa giản đồ mã hóa ký tự
- ✓ **errorPage:** Định nghĩa URL của JSP khác mà báo cáo trên Java các unchecked runtime exception.
- ✓ **isErrorPage:** Chỉ rằng nếu JSP page này là một URL được xác định bởi thuộc tính ErrorPage của JSP page khác
- ✓ **extends:** Xác định một lớp cha mà Servlet đã tạo phải kế thừa
- ✓ **import:** Xác định một danh sách các package hoặc class để sử dụng trong JSP khi lệnh import trong Java làm việc cho các lớp Java
- ✓ **isThreadSafe:** Định nghĩa mô hình thread

Thẻ chỉ dẫn

Jsp Include Directive, cú pháp:

```
<%@ include file="resourceName" %>
```

JSP **Taglib** directive, cú pháp:

```
%@ taglib uri="uriofthetaglibrary" prefix="prefixoftaglibrary" %>
```

Thẻ JSTL

Biểu thức: **`${expr}`**

<https://www.javatpoint.com/jstl-core-tags>

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
<html>
<head>
<title>Core Tag Example</title>
</head>
<body>
<c:forEach var="j" begin="1" end="3">
    Item <c:out value="${j}"/><p>
</c:forEach>
</body>
</html>
```

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
<html>
<head>
<title>Core Tag Example</title>
</head>
<body>
<c:set var="income" scope="session" value="${4000*4}"/>
<c:if test="${income > 8000}">
    <p>My income is: <c:out value="${income}"/><p>
</c:if>
</body>
</html>
```

JSTL SQL Tags List

```
<%@ page import="java.io.*,java.util.*,java.sql.*"%>
<%@ page import="javax.servlet.http.*,javax.servlet.*" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>

<html>
<body>

<sql:setDataSource var="db" driver="com.mysql.jdbc.Driver"
    url="jdbc:mysql://localhost/test"
    user="root" password="root"/>

<sql:query dataSource="${db}" var="rs">
SELECT * from Student;
</sql:query>

<table border="2" width="100%">
<tr>
<th>Student ID</th>
<th> Name</th>
<th> Address</th>
<th> Mark</th>
</tr>

<c:forEach var="table" items="${rs.rows}">
<tr>
<td><c:out value="${table.id}"/></td>
<td><c:out value="${table.name}"/></td>
<td><c:out value="${table.address}"/></td>
<td><c:out value="${table.mark}"/></td>
</tr>
</c:forEach>

</table>
</body>
</html>
```

Thẻ hành động

JSP Action Tags	Description
<code>jsp:forward</code>	Điều chuyển đến một tài nguyên khác
<code>jsp:include</code>	includes another resource.
<code>jsp:useBean</code>	creates or locates bean object.
<code>jsp:setProperty</code>	sets the value of property in bean object.
<code>jsp:getProperty</code>	prints the value of property of the bean.
<code>jsp:param</code>	sets the parameter value. It is used in forward and include mostly.

jsp:useBean action tag

```
<jsp:useBean id= "instanceName"  
    scope= "page | request | session | application"  
    class= "packageName.className" >  
</jsp:useBean>
```

```
public class Calculator{  
  
    public int cube(int n){return n*n*n;}  
  
}
```

index.jsp file

```
<jsp:useBean id="obj" class="com.javatpoint.Calculator"/>  
  
<%  
    int m=obj.cube(5);  
    out.print("cube of 5 is "+m);  
    %>
```


jsp:setProperty action tag

Syntax of jsp:setProperty action tag

```
<jsp:setProperty name="instanceOfBean" property="*" |  
property="propertyName" param="parameterName" |  
property="propertyName" value="{ string | <%= expression %>}"  
/>
```

Example of jsp:setProperty action tag if you have to set all the values of incoming request in the bean

```
<jsp:setProperty name="bean" property="*" />
```

Example of jsp:setProperty action tag if you have to set value of the incoming specific property

```
<jsp:setProperty name="bean" property="username" />
```

Example of jsp:setProperty action tag if you have to set a specific value in the property

```
<jsp:setProperty name="bean" property="username" value="Kumar" />
```

jsp:getProperty action tag

Syntax of jsp:getProperty action tag

```
<jsp:getProperty name="instanceOfBean" property="propertyName" />
```

index.html

```
<form action="process.jsp" method="post">  
Name:<input type="text" name="name"><br>  
Password:<input type="password" name="password"><br>  
Email:<input type="text" name="email"><br>  
<input type="submit" value="register">  
</form>
```

process.jsp

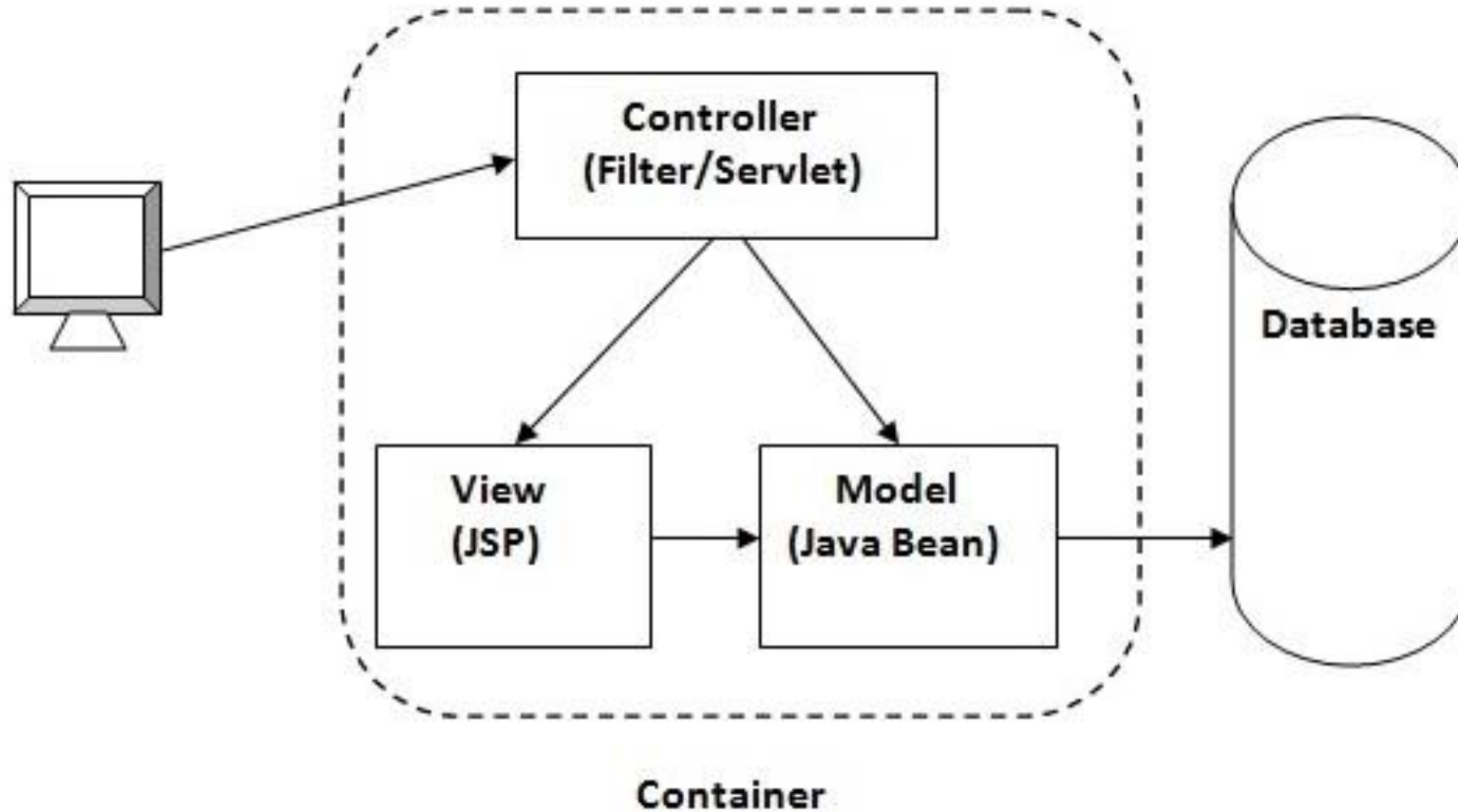
```
<jsp:useBean id="u" class="org.sssit.User"></jsp:useBean>  
<jsp:setProperty property="*" name="u"/>  
  
Record:<br>  
<jsp:getProperty property="name" name="u"/><br>  
<jsp:getProperty property="password" name="u"/><br>  
<jsp:getProperty property="email" name="u" /><br>
```

User.java

```
package org.sssit;  
  
public class User {  
    private String name,password,email;  
    //setters and getters  
}
```

<https://www.javatpoint.com/jsp-setProperty-and-jsp-getProperty-action-tag>

MVC in JSP



Ví dụ MVC sử dụng JSP

index.jsp một trang để nhập user, password;

LoginBean.java chứa dữ liệu

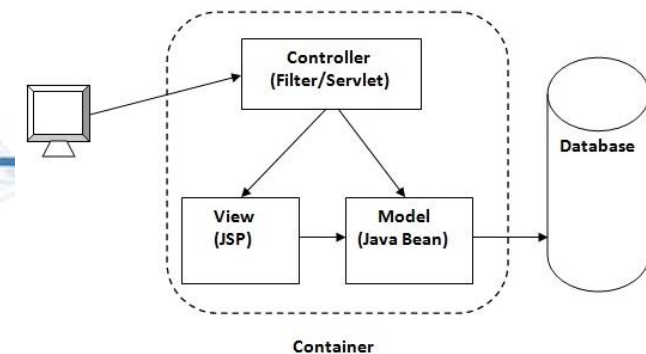
ControllerServlet.java xử lý logic:

Login thành công -> **login-success.jsp**

Login không thành công -> **login-error.jsp**

Tham khảo: <https://www.javatpoint.com/MVC-in-jsp>

Ví dụ MVC sử dụng JSP



File: index.jsp

```
<form action="/ControllerServlet" method="post">
Name:<input type="text" name="name"> <br>
Password:<input type="password" name="password"> <br>
<input type="submit" value="login">
</form>
```

```
public class LoginBean {
private String name,password;
public String getName() {
return name;
}
public void setName(String name) {
this.name = name;
}
public String getPassword() {
return password;
}
public void setPassword(String password) {
this.password = password;
}
public boolean validate(){
if(password.equals("admin")){
return true;
}
else{
return false;
}
}
```

```
@WebServlet("/ControllerServlet")
public class ControllerServlet extends HttpServlet {
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
response.setContentType("text/html");
PrintWriter out=response.getWriter();
String name=request.getParameter("name");
String password=request.getParameter("password");
LoginBean bean=new LoginBean();
bean.setName(name);
bean.setPassword(password);
request.setAttribute("bean",bean);
boolean status=bean.validate();
if(status){
RequestDispatcher rd=request.getRequestDispatcher("login-success.jsp");
rd.forward(request, response);
}
else{
RequestDispatcher rd=request.getRequestDispatcher("login-error.jsp");
rd.forward(request, response);
}
}
@Override
protected void doGet(HttpServletRequest req, HttpServletResponse resp)
throws ServletException, IOException {
doPost(req, resp);
}
```

Ví dụ MVC sử dụng JSP

```
//File: login-success.jsp
<%@page import="mta.LoginBean"%>

<p>You are successfully logged in!</p>
<%
LoginBean bean=(LoginBean) request.getAttribute("bean");
out.print("Welcome, "+bean.getName());
%>
//File: login-error.jsp
<p>Sorry! username or password error</p>
<%@ include file="index.jsp" %>
```

Xây dựng ứng dụng minh họa

1. Xây dựng ứng dụng thao tác với CSDL

<https://www.javatpoint.com/crud-in-jsp>

2. Xây dựng ứng dụng E-Commerce

Tham khảo Book 2 - Chapter 18. Developing E-Commerce Applications